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Try the interactive case study live: bsilva87921.github.io/loan-rules-lab

These case studies show how I deliver enterprise initiatives: turning policy into clear business rules, proving them with real data, and getting operations ready before go-live.

IMPACT AT A GLANCE

14 yrs

At JPMorgan Chase, from the front line to Senior Business Analyst

\$20M

Saved by pinpointing the true driver of rising handle time

30,000

Employees across 5 lines of business on scorecards I governed

9,578

Real loans used to test lending rules end to end

HOW MY WORK MAPS TO THE ROLE

WHAT THE ROLE NEEDS	EVIDENCE IN THESE CASE STUDIES	PAGE
 Enterprise initiatives Partnering with business, technology and operations	Scorecard program for 30,000 employees across 5 lines of business , delivered with product, operations and finance partners.	3
 Requirements & business rules Policy into rules a system can run	Numbered business rules for a loan decision flow, and a 94.4% reconstruction of current credit policy from the data.	2
 Testing & UAT Test cases, acceptance criteria, defects	Test cases built from real loans, an acceptance test with a severity-ranked defect log, and UAT on every new or changed metric at Chase.	2-3
 Analysis & decision support Root cause, forecasting, recommendations	Root-cause analysis that saved \$20M ; a cross-validated risk model; ranked, goal-based recommendations for leaders.	2-3
 Operational readiness Staffing, process change, automation	Review-workload and staffing forecasts before go-live; automation saving 0.5 FTE ; former Operations Manager of 50+ associates.	2-4

Rolling out new loan decision rules

The challenge: when a bank launches a new loan origination process, the business analyst has to turn credit policy into rules a system can run, prove the rules work before go-live, and tell operations how much manual review to staff for. I built the **Loan Rules Lab** to do that end to end on **9,578 real LendingClub loans** (2007–2010, Kaggle).

1 DISCOVER

Validated the data, then rebuilt current policy: **3 rules match it on 94.4%** of loans.

2 DEFINE

Wrote numbered business rules (BR-01...), each with how many loans it affects.

3 TEST

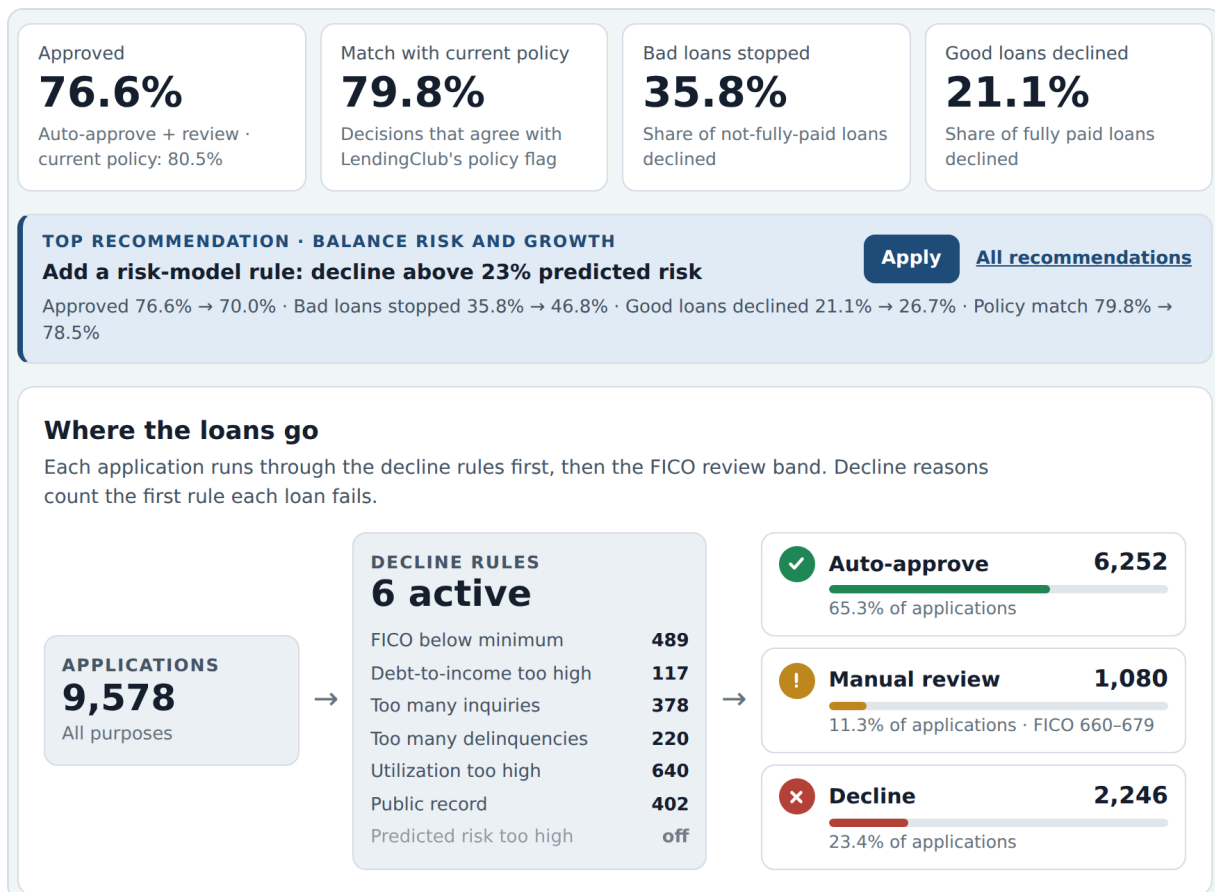
Built test cases from real loans, plus an acceptance test with a **severity-ranked defect log**.

4 PREDICT

Trained a cross-validated risk model: AUC **0.67** vs. **0.62** for FICO alone.

5 READY

Forecast review workload and staffing; ranked changes by goal for leaders.



The lab: live results, the top recommendation, and where every loan goes as the rules change.

94.4%

Match to current policy with 3 rules (approve-all baseline: 80.5%)

42% vs 36%

Bad loans stopped: model vs. fixed rules, at the same cost

1,932

Policy mismatches logged as defects by severity (Balanced rules)

0.11 FTE

Review staffing at 400 applications a month

In a new role: the same steps apply to any loan platform rollout: document the decision rules, prove them with real scenarios, log defects by severity, and size the review team before go-live.

Methodology: data checked in Python; logistic regression with 5-fold cross-validation (scikit-learn); every number recalculated live in a self-contained web app. Live: bsilva87921.github.io/loan-rules-lab

Delivering results across the enterprise

02 Pinpointing the true driver of rising handle time

Senior Business Analyst

Root-Cause Analysis • Data Analysis • Financial Impact

STARTING POINT

Average handle time (AHT) rose about **5 seconds** across the business.



WHAT THE DATA SHOWED

Hiring sprints had grown the new-hire pool. Associates past 6 months were **20 seconds** faster.



ACTION

Targeted team-leader coaching for associates under 6 months, plus a call-type dashboard.

Handle time reversed course

Change in average handle time, seconds

Initial rise

After coaching and dashboard



\$20M

Total savings across card and retail

30 sec

AHT reduction

In a new role: the same discipline, finding the real driver before prescribing a fix, applies when a new process underperforms after launch.

03 Scorecard governance for 30,000 employees

Senior Business Analyst | Relationship Manager

Requirements • Data Dictionaries • UAT • Automation

DEFINE

KPIs and business rules with leaders

DOCUMENT

Metric definitions and data dictionaries

VALIDATE

UAT on every new or changed metric

DELIVER

SQL, Power BI and Alteryx automation

RECALIBRATE

Monitor and realign to business goals

25%

More accurate reporting

30%

Less operational friction within 6 months

0.5 FTE

Saved by automating data workflows

In a new role: clear definitions and tested changes are what keep a bank-wide initiative audit-ready and trusted by every partner.

HOW I DELIVER

From idea to go-live on a strategic initiative

PHASE	WHAT I DELIVER	WHERE I'VE DONE IT
Discover	Current-state process maps, data review, stakeholder goals	Rebuilt current credit policy from loan data (94.4% match)
Define	Requirements, business rules, acceptance criteria	Metric definitions and data dictionaries for 5 lines of business
Test	Test plans, test cases, defect logs by severity	UAT on every scorecard change; test cases built from real loans
Decide	Options with trade-offs and a clear recommendation	Root cause behind a \$20M saving; ranked rule recommendations
Launch	Operational readiness, training, staffing forecasts	Led 50+ associates through process change; review-staffing forecast
Measure	Baselines, KPIs, post-launch tracking and automation	Scorecards for 30,000 employees; automation saving 0.5 FTE

ID	SCENARIO	GIVEN	EXPECTED	ACTUAL	RESULT
TC-001	Decline when FICO is below 660	LC-7718: FICO 657, DTI 19.3%, 0 inquiries, 0 delinquencies, utilization 77.3%, no public records	Decline	 Decline	Pass
TC-002	Decline when debt-to-income is above 25%	LC-8088: FICO 757, DTI 25.1%, 0 inquiries, 0 delinquencies, utilization 50.8%, no public records	Decline	 Decline	Pass
TC-003	Decline when inquiries exceed 5	LC-0361: FICO 777, DTI 2.1%, 6 inquiries, 0 delinquencies, utilization 14.3%, no public records	Decline	 Decline	Pass
TC-004	Decline when delinquencies exceed 1	LC-0026: FICO 712, DTI 10.0%, 0 inquiries, 2 delinquencies, utilization 36.6%, no public records	Decline	 Decline	Pass

Deliverable example: test cases built from real loans, each with expected result, actual result and pass/fail.

MORE WORK

Live projects: [Fundraising Analytics Dashboard](#) (1M+ gift records) • [Probable: Live Sports Lab](#) (4 probability models) • [Alteryx Workflow Lab](#) (24 guided steps)

I turn strategy into clear requirements, prove the change works, and make sure the business is ready for it.

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